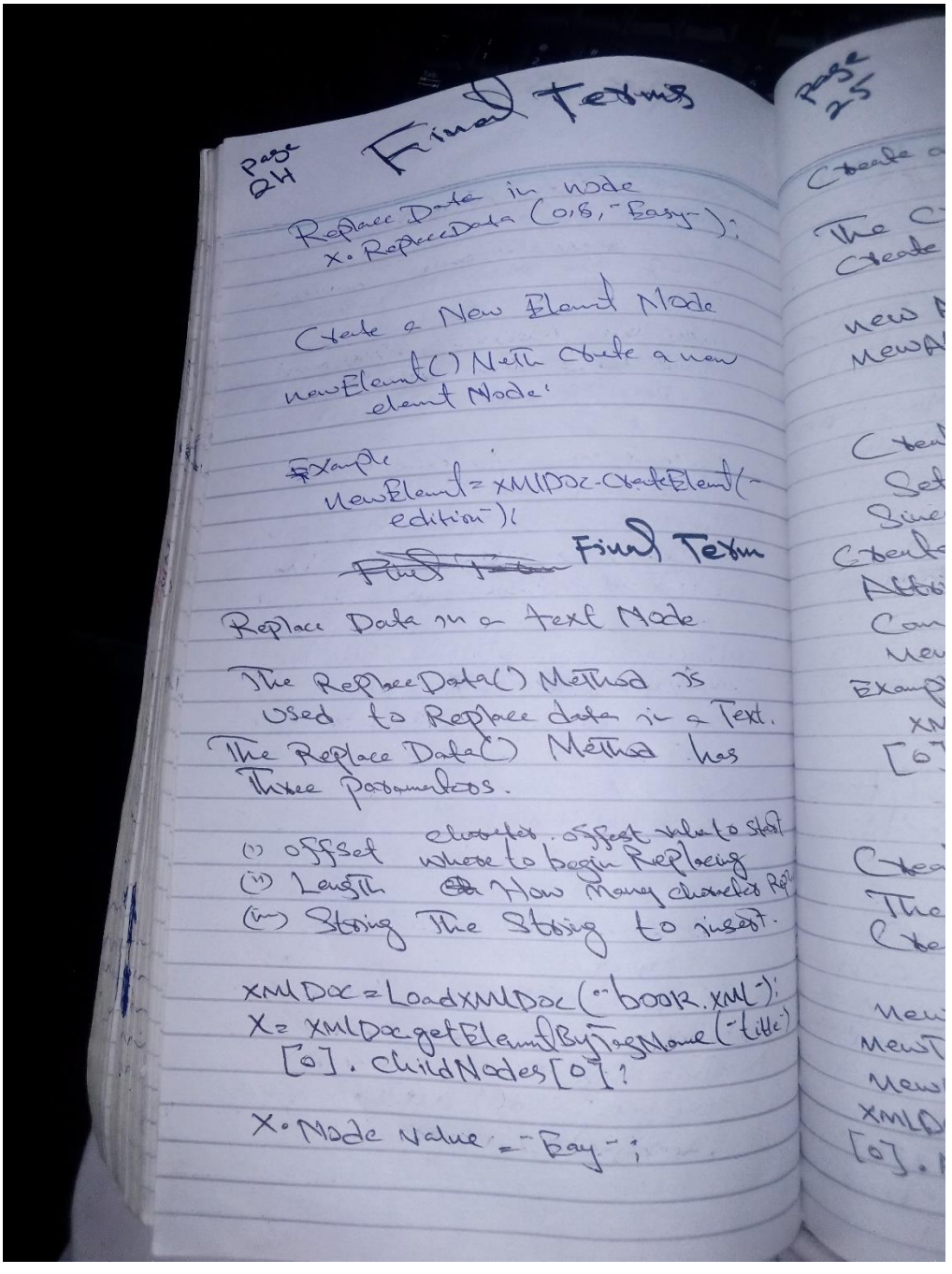
Final Term
Handwritten Short Notes: Fall 2024
CS311 – introduction to web services

MADE BY MUHAMMAD AYAZ



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Create a New Attribute Node

The CreateAttribute() is used to create a new Attribute node

```
newAtt = xmlDoc.CreateAttribute("edition");
newAtt.nodeValue = "first";
```

Create an Attribute using SetAttribute()

Since The SetAttribute() Method Create new Attribute if the Attribute does not exist. it can be used to Create a new Attribute

Example

```
xmlDoc.getElementsByTagName("book")[0].SetAttribute("edition", "first");
```

Create Text Node

The CreateTextNode() Method Create a new text node

```
newEle = xmlDoc.CreateElement("edition");
newText = xmlDoc.CreateTextNode("first");
newEle.AppendChild(newText);
xmlDoc.getElementsByTagName("book")[0].AppendChild(newEle);
```

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Create a CDATA Section Node
The CreateCDATASection()
Method Create a new CDATA
node.

```
newCDATA = XMLDoc.CreateCDATASection
("Special offer and Book Sale");
XMLDoc.getElementByTagName("book")
[0].appendChild(newCDATA);
```

Create a Comment Node
The CreateComment() Method a
new Comment Node

```
newComment = XMLDoc.CreateComment
("Revised March 2015");
XMLDoc.getElementByTagName("book")
[0].appendChild(newComment);
```

Add a Node AppendChild()
The AppendChild() Method add
a child node to an existing node.
The new node is Added (appended)
after Any exisiting Child nod.

```
newEle = XMLDoc.CreateElement("edition")
XMLDoc.getElementByTagName("book")
[0].appendChild(newEle);
```

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Append The text node to The <addition> node
Append The <addition> node to The <books> elemt.

insert a node insertBefore()
The insertBefore() Method insert a node before is Specified Child node. This Method is useful when The position of The Added node is important.

```
newNode = XMLDoc.CreateElement("book")

x= xmlDoc.documentElement;
y= xmlDoc.getElementByTagName("book")
   [3];
x.insertBefore(newNode,y);
```

Add a New Attribute
The SetAttribute() Method Sets The value of an Attribute.

```
XMLDoc.getElementByTagName("book")[0]
.SetAttribute("edition","first");
```

The SetAttribute() Method will overwrite The existing value.

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Add Text to a Node - insertData()

The insertData() Method inserts data into an existing textNode. The insertData() Method has two parameters:

offset - where to begin inserting characters (start at Zero)

String - The String to insert.

```
XmlDoc.getElementByTagName("title")
[0].childNodes[0].insertData
(0,"Easy");
```

Copy a Node

The cloneNode() Method Creates a copy of a Specified node.

The cloneNode() Method has a parameter (true or false). This parameter indicates if the cloned node Should include All Attributes And child nodes of original node.

```
oldNode = XmlDoc.getElementByTag
Name("book")[0];
newNode = oldNode.cloneNode(true);
XmlDoc.documentElement.AppendChild
(newNode);
```

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Topic 168

JAXP Java API for XML processing

JAXP is used to Read the XML documents or parse the XML documents

JAXP API provides classes And interfaces to create parser Software in a java Application.

The Parser Software is used to Read, validate and parse the XML documents.

Three Types of parser's

(1) Dom parser (Document object Model)
(3) SAX parser (Simple API for XML)
(3) Stack parser (Streaming API for XML)

JAXP also supports the Extensible Stylesheet Language Transformation

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The SAX And DOM API are defined by The XML Dev group And by The W3C, Respectively. The Libraries That define Those API are as follows:

Java.xml.Parsers:
Provide a Common interface for different vendor's SAX And DOX parsers.

Org.w3c.dom: Define The Document class (a DOM) as well as Classes for All The Components of Dom.

org.xml.sax: Define The Basic SAX API.

Java.xml.transform:
Define The XSLT That Let you transform XML into other forms.

Java.xml Stream:
Provide StAX Specific transform APIs.

SAX
The
is
Mech
elem
This
to
The

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SAX Package

The Simple API for XML (SAX) is the event driven, Serial Mechanism that does element by element processing. The API for this Level Reads and write XML to a data ~~Rep~~ Repository or the web

Dom Packages

The Dom API is Generally an easier API to use. It provide a familiar tree structure of objects. you can use the Dom API to manipulate the hierarchy of application objects it encapsulates.

The Dom API is ideal for interactive application because the entire Model is present in Memory

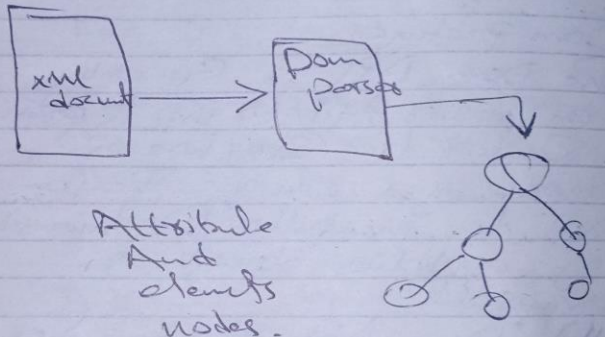
Comparison

Dom Requires Reading the entire XML Structure And holding the object tree in Memory, So it Much more CPU And Memory intensive. For that Reason the SAX API tends to be preferred for Server Side Application and data filter that

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Dom Parser
Document object Model
parser is a S/W used to
Read, Validate and Parse
The XML document

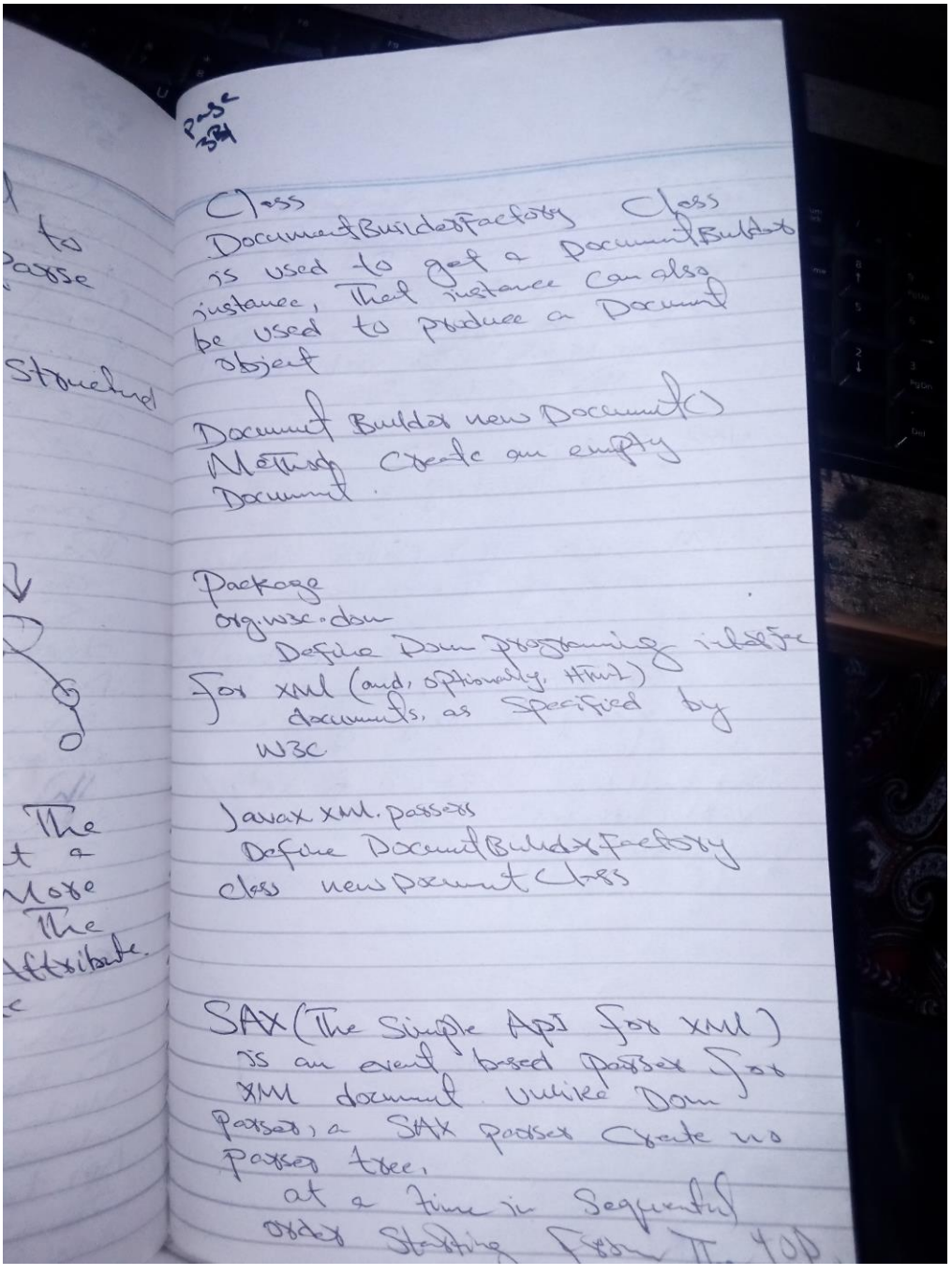
Dom parser is a tree Structured
Based Parsers.



Attributes
And
elements
nodes.

Dom Parser will Read The
entire XML document at a
time and Consuming More
memory to Represent The
XML elements And Attributes.
Support Read write
Sequence Manner
Random Manner.
is Slow

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Class
DocumentBuilderFactory Class
is used to get a DocumentBuilder
instance, That instance can also
be used to produce a Document
object

Document Builder new Document()
Method Create an empty
Document.

Package
org.w3c.dom
Define Dom programing interface
for XML (and, optionally, HTML)
documents, as specified by
W3C

Javax XML.passers
Define DocumentBuilderFactory
class new Document Class

SAX (The Simple Api for XML)
is an event based parser for
XML document Unlike Dom
parser, a SAX parser Create no
parse tree,
at a time is Sequential
order Starting From T. to D.

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Content Handler interface
This interface Specifies the Callback Method that the SAX Parser uses to notify an Application program of the components of the XML document that it has Seen.

void startDocument()
Called the beginning of a document

Void endDocument() Called at the end of the document.

Void startElement(String uri, String LocalName, String qname, Attributes atts)
Called the beginning of an element.

Void endElement Called End.
void Characters Characters data
void ignorableWhitespace

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JDom Java Document object Model is an open source Java based library to parse XML Document And it is Typically Java developer friendly API. Lightweight and fast, and is optimised for Java developer.
combine Dom And SAX

Advantages
JDom gives Java developers flexibility and easy Maintainblity of XML parsing code. it is Light weight and quick API.

JDom Classes
Document. Represent The entire doc.
Element. Represent Element.
Attribute Represent Attribute.
Text Represent Text.
Comment Represent comment.

Steps to using JDom
(i) Import XML Related packages.
(ii) Create a SAX Builder
(iii) Create a Document from a file or Stream.
(iv) Extract The Root element.
(v) Examine Attribute
(vi) Examine Sub elements.

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Import xml Related Packages
(i) import java.io.*;
(ii) import java.util.*;
(iii) import org.jdom2.*;

Create a documentBuilder

SAXBuilder Sax = new SAXBuilder();

Create a document From a File Stream.

File inputFile = new File("input.Text")

SAXBuilder Sax = new SAXBuilder();

Document document = SaxBuilder.build(inputFile

Extract The Root element
Element classElement = document.getRootElement

Examine Attribute
getAttribute("AttributeName");

Examine Sub element
getChildren();

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JDom XML parser Modify an existing XML file.

(i) Add new elemt.
(ii) Update existing elemt Attribute.
(iii) Update existing elemt value
(iv) Delete existing elemt.

Example

Add new Age

Updated The Step Attribute id=2

Update Safety vale to 7.00

Delete first name.

JAXB

Java Architecture For XML Binding

This API is used convert Java object into XML and XML into Java object.

Converting of java object into XML is called Marshling

convert XML to java unmarshling

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Validation
is the process of verifying that an XML document meet all the constraint expressed in the schema. JAXB 1.0 provided validation at unmarshal time and also enabled on demand validation on a JAXB content tree.

XML Binding
is the process of representing information in XML document as an object in computer memory. XML Data Binding application access XML Data direct from the object instead of using the Document object Model.

Annotation
An Annotation is extra information associated with particular point in a document or that piece of information.

Java Annotations for Augmenting the generated classes with Additional information

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Module 15

Servlets

Servlet Technology is used to create web Application (Reside at Server Side And generates dynamic web page.

There are many interfaces and classes in the Servlet API such as Servlet, GenericServlet, HttpServlet, ServletRequest, ServletResponse etc.

Web Application

Application which Accessible from the web
Web Application is composed of web components like Servlet JSP, Filter etc.
Respond HTTP ~~Get~~ And other Request.

CGI (common Gateway interface)
CGI Technology enables the web Server to call an external program And pass HTTP Request information to the external program to process the Request.

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HTTP (Hyper Text Transfer Protocol)

HTTP is the protocol that allows web servers and browsers to exchange data over the web.

It is Request and Response protocol

Http Request Method
Get
And post mostly use

A Servelt Life Cycle can be defined as the entire process from its Creation till the destruction

(1) Servelt initialized by init() Method
process Client Request Service()
Terminated by destroy() Method
garbage Collect from JVM Garbage

doGet() Method
Request Result from a normal Request for a URL or from HTML form that has no Method.

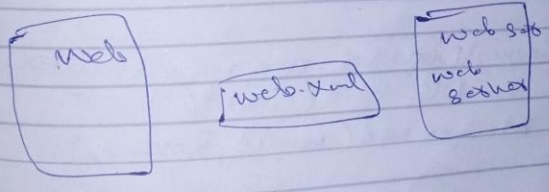
doPost() Method
Form the post Html Request handler

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Create a Servelt steps
(i) By implementing The Servlet interface
(ii) By inheriting The Generic Servlt Class.
(iii) By inherit The HTTPServelt Class.

```
import java.servlet.http*;
import javx.servlet.*;
import java.io*;
```

web.xml (deployment dispriptor) file defines mappings between URL path and The Servets That handle Request with Those path



web.xml file Data

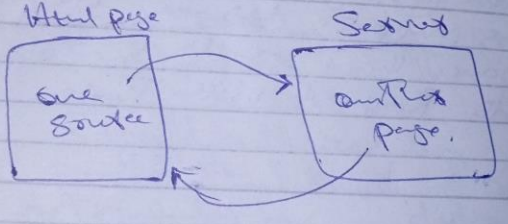
```
<web.app>
<Servelet-mapping>
<Servelet-name>
<URL-pattern>
<Servelt_Mapping>
```

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```
<Servelt>
    <Servelt-Name>
    <Servelt-Class>
</Servelt>
</web.APP>
```

RequestDispatcher in Servlet

The RequestDispatcher interface provide the facility of dispatching the Request to another Resource it may be html, Servlet or jsp.




```
RequestDispatcher rd = request.getRequest
    Dispatcher("Servlet2");
rd.forward(request, Respance);
```

Servelt Context interface

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The object of ServletContext provide an interface between the container and Servlet. ServletContext object can be used to get configuration information.

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Servlet context interface

object of ServletContext is created by the web container at time of deploying the project. This used to get information from web.xml file. There is only one object per web Application

Advantages

Easy to maintain if any information is shared to All the Servlet, it is better to make it available for All the Servlets.

Use of Servlet Context

(1) provide interface between the Container and Servlet.
(2) Used to get configuration information from web.xml file.
(3) Used to get Set Remove Attribute from web.xml file.
(4) Used to provide intra Application Communication.

Method of Servlet context

getInitParameter(Return value of Specified Parameter name.
Set Attribute()
get Attribute()
Remove Attribute()

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Session Tracking Servlets

Session Simply Means a particular interval of time what

There are Four techniques used in Session tracking

(1) Cookies
(2) Hidden Form Field
(3) URI Rewriting
(4) HttpSession

Cookies

is Small Piece of information That is persisted between the Multiple Client Requests

Cookies work

By default, each Request is Considered as a new request. In Cookies Technique, we add Cookie with Response from The Servlet. So cookies is Stored in Cache of The browser. After That if request is Sent by The user, Cookie is added with request by default. Thus, we Recognize The user id user.

Two Types of cookies

Non persistent cookie
it is valid for Single Session only
it is Removed each time when user

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Persistent cookie
it is valid for Multiple Session
it is not removed each time
when close the browser it is
removed only if user Logout or
Signout.

Advantages of cookies
Simplest Technique of maiting
the State.
cookies are maintained at Client side

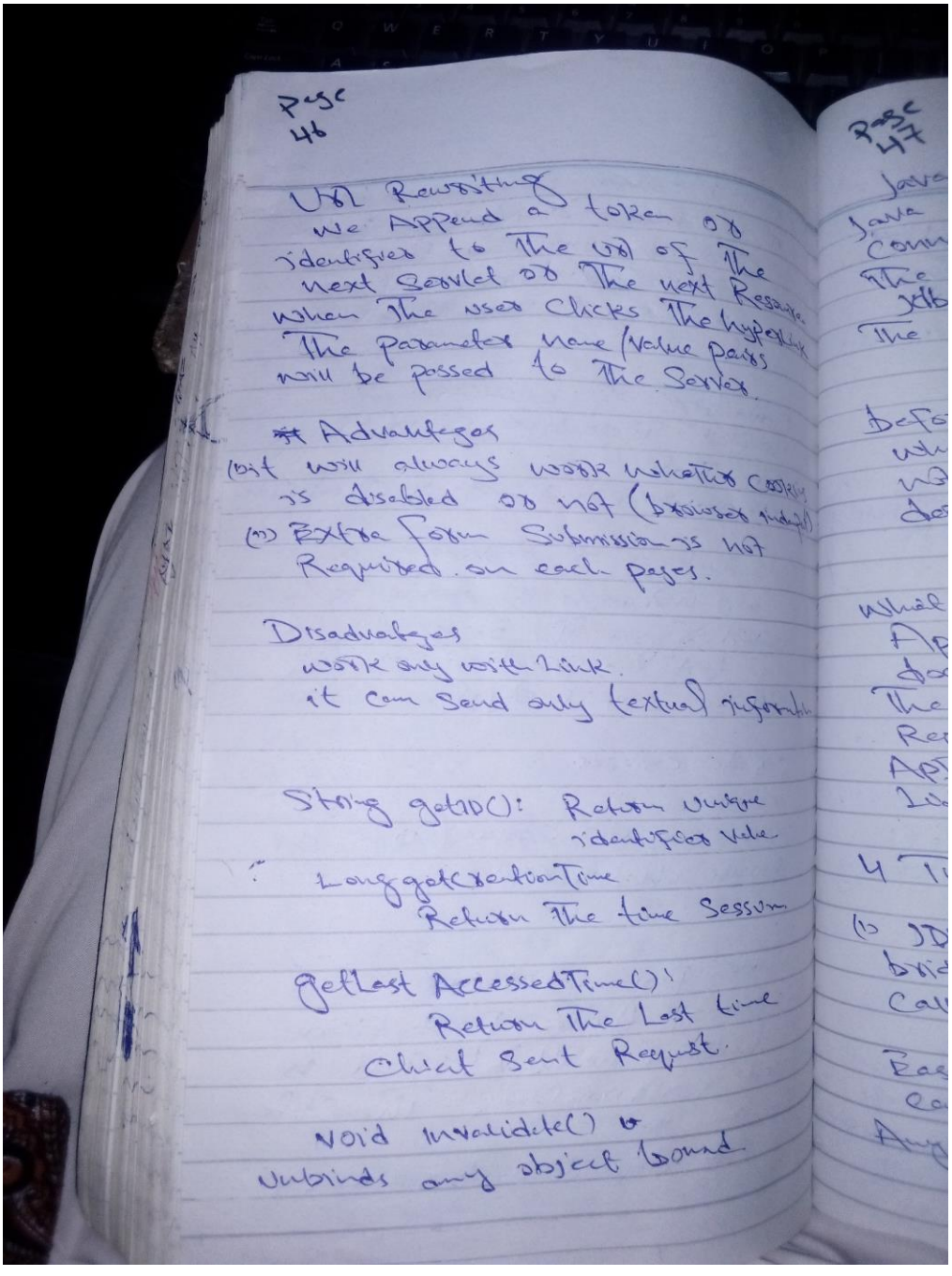
Disadvantages of cookies
it will not work if cookie is
disabled from the browser.
only textual information can be
Set in cookies object.

Create cookies

```
Cookie ck = new Cookie("user");
response.addCookie(ck);
```

Hidden Form field
in case of Hidden Form field
a hidden (invisible) textfield is
used for maintaining the State
of an user
Not depend on browser

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URL Rewriting
We Append a token or identifier to the URL of the next Servlet or the next Resource. when the user clicks the hyperlink The parameter name/value pairs will be passed to the Server.

* Advantages
(i) it will always work whether cookie is disabled or not (browser independent)
(ii) Extra form Submission is not Required on each pages.

Disadvantages
work only with Link.
it can Send only textual information.

String getID(): Return Unique identifier value

Long getCreationTime
Return the time Session

GetLastAccessedTime():
Return the Last time Client Sent Request.

void invalidate() u
unbinds any object bound.

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Java
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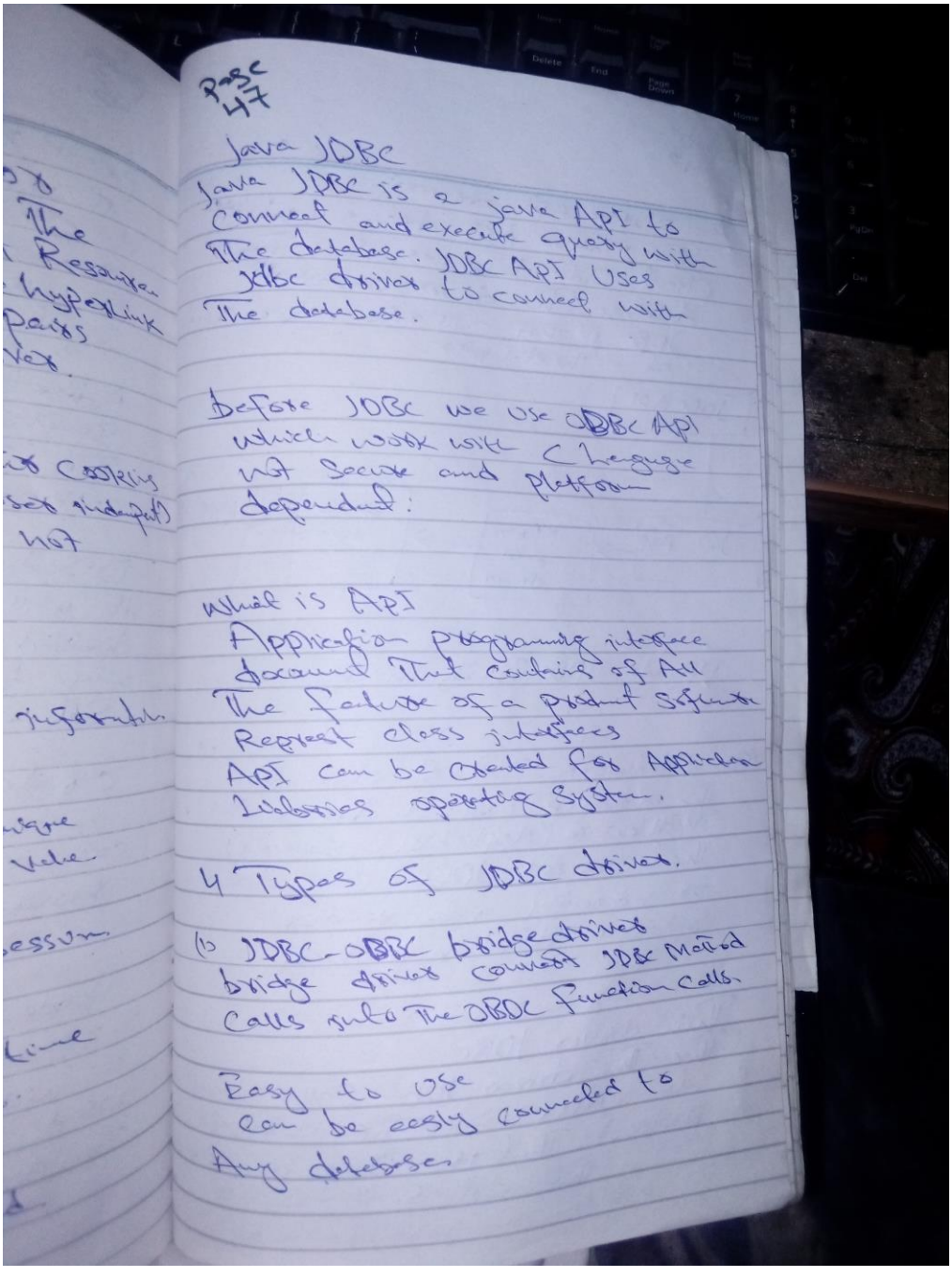
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Java JDBC

Java JDBC is a java API to connect and execute query with the database. JDBC API uses JDBC driver to connect with the database.

Before JDBC we use ODBC API which was written in C language and platform dependent.

What is API

Application programming interface document that contains of All the feature of a product software Repeat class interfaces API can be created for Applications libraries operating System.

4 Types of JDBC drivers.

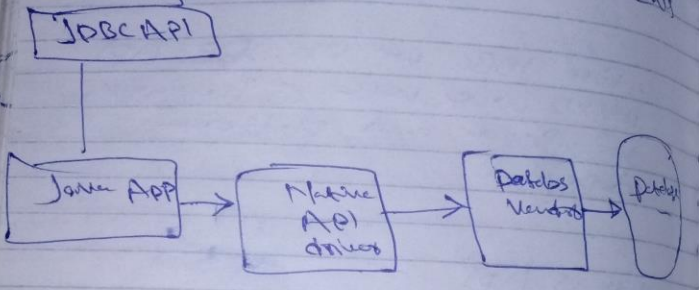
(1) JDBC-ODBC bridge driver
bridge driver convert JDBC Method calls into the ODBC function calls.

Easy to use
can be easily connected to
Any databases

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2 Native API driver

The Native API driver uses the Client Side Libraries of the database. The driver converts JDBC Method calls into native calls of the database.



Upgrade JDBC Bridge driver.

(3) Network protocol driver.

The Network protocol driver uses Middleware (Application Server) that converts JDBC calls directly or indirectly into vendor specific database protocol.

Thin driver JDBC

The Thin drivers JDBC Calls directly into vendor Specific database protocol. That is why it known as Thin driver. it is fully written in Java Langage.

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5 Steps to connect the database in Java

(i) Register the driver class forName()
(ii) Creating connection getConnection()
(iii) Creating statement createStatement()
(iv) Executing queries executeQuery()
(v) Closing connection close()

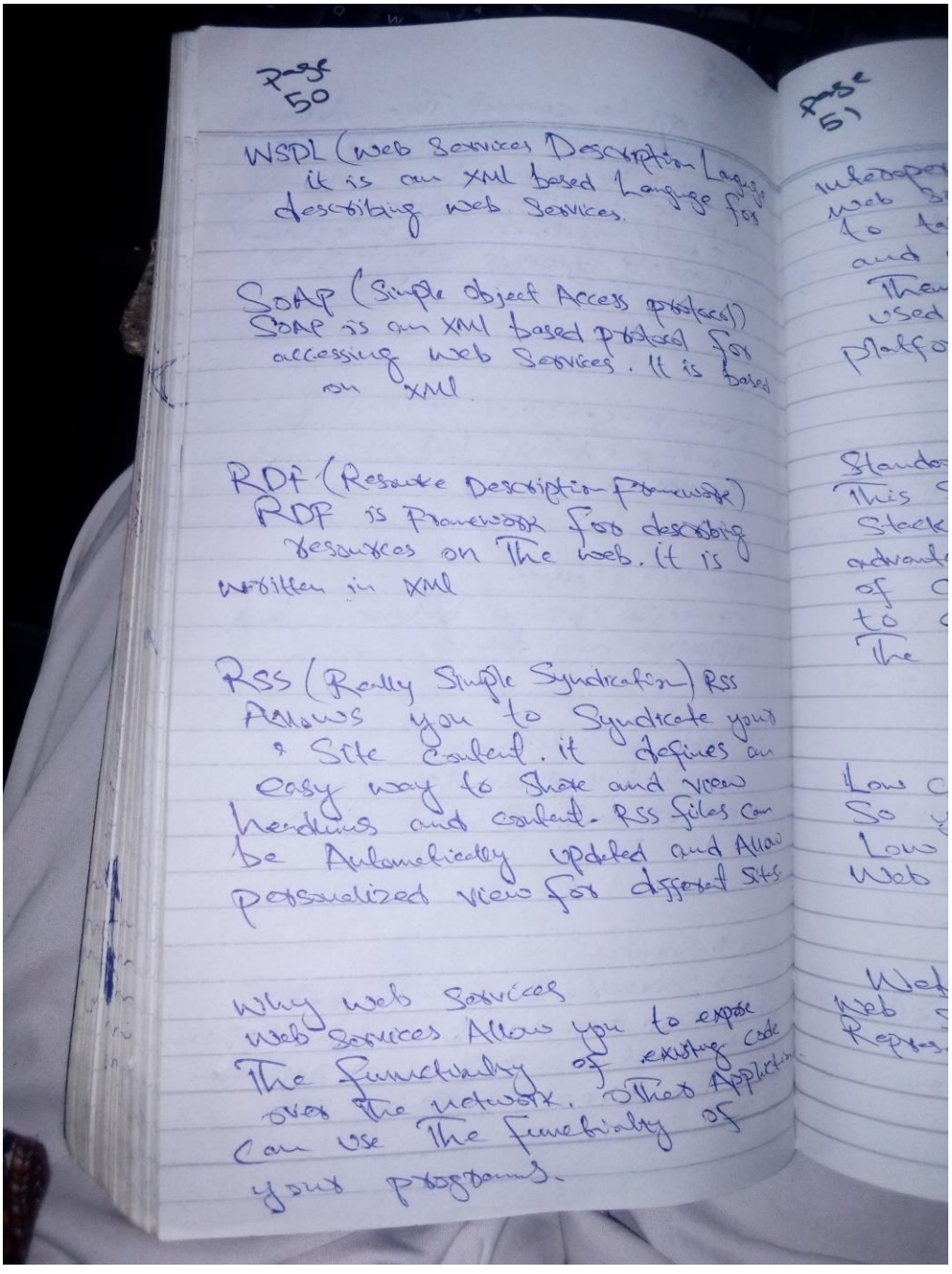
Driver Manager Class is used for creating establish between driver in database.

Connection interface
connection session between java Application and database.

Statement interface
provides Methods to execute queries with the database

Web Services
web services are open standard (xml, SOAP, HTTP etc) based web Applications that interact with other web Application for the purpose of exching data.

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WSDL (Web Services Description Language)
it is an XML based Language for describing web Services.

SOAP (Simple Object Access protocol)
SOAP is an XML based protocol for accessing web Services. It is based on XML.

RDF (Resource Description Framework)
RDF is Framework for describing resources on the web. it is written in XML

RSS (Really Simple Syndication) RSS Allows you to Syndicate your Site Content. it defines an easy way to Share and view headlines and content. RSS files can be Automatically updated and Allow personalized view for different Sites

Why web Services
web Services Allow you to expose the functionality of existing code over the network. Other Application can use the functionality of your programs.

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Interoperability
Web Services Allow Various Applications to talk to each other and share data and Services Among themselves. Web Services are used to make the Application platform and Technology independent.

Standardized protocol
This Standardization of protocol Stack gives the business many advantages such as a wide range of choices reduction in the cost to competition and increase in the quality.

Low cost of Communication
So you can use your existing Low cost internet for implementing Web Services.

Web Services Characteristics
Web Services uses XML at data Representation and Transport Layer.

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Loosely coupled
Adopting a Loosely coupled Architecture tends to make Software System More Manageable and Allow Simpler integration between different Systems.

Coarse Grained
Web Services Technology provide a natural way of defining Coarse grained Services that Access the right Amount of Business Logic.

Ability to be Synchronous or Asynchronous
Client Behaviour Three result at at and at a time

Support Remote procedure Calls (RPCs)
Allow Client to invoke Function or Remote Object on XML based protocol.
Java Beans and .NET components More enhance this Technology

Support Document Exchange

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XM
This is
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SOAP
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WSDL
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for
A

UDD

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XML RPC

This is the simplest XML based protocol for exchanging information between computers.

SOAP

SOAP is an XML based protocol for exchanging information between computers.

WSDL

WSDL is an XML based language for describing web services and how to access them.

UDDI (Universal Description Discovery Integration)

UDDI is an XML based standard for describing, publishing and finding web services.

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Web Services Architecture

Web Services Roles

Service provider
Provider of web services.

Services Requestor
Any consumer of the web service
Send Request for Web Services.

Services Registory
The registory provides a central
place where developers can publish
new services or find existing ones.

Web Services protocol Stack
A second options of viewing
the webservices.

Service Transport
This Layer is responsible for
transporting Messages between
Application (HTTP, SMTP, FTP, BEEP)

(1) **XML Messaging**
This Layer is Responsible for
encoding Messages in a common
XML format.

(2) **Services Description**
For Describing

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